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Paleomagnetism of the Hart Dolerite (Kimberley, Western Australia) – A two-stage assembly of the supercontinent Nuna?

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ABSTRACT

We present new paleo- and rock magnetic results from the ca. 1792 Ma Hart Dolerite sills that intrude the strata of the Kimberley craton, Western Australia. From 24 sites sampled, 23 are directionally clustered and the site mean directions were used to calculate a grand mean direction. Ten of the 23 sites have 95% confidence intervals (α_{95}) less than 16°. A positive reversal test, dissimilarity of the corresponding paleopole from known younger Australian poles, and a previously reported positive baked contact test, indicate that our paleomagnetic results are of primary origin, and the results are consistent with the geologic interpretation that the Hart Dolerite sills represent a multiphase magmatic emplacement over the duration of several million years. The new ca. 1792 Ma paleopole for the Kimberley craton is applicable to the entire North Australian Craton. A comparison of Paleoproterozoic poles from the North and West Australian cratons supports the occurrence of a previously proposed 40° intracontinental rotation between the two cratons during latest Neoproterozoic time. Furthermore, comparing these Australian poles with similar aged Laurentian poles shows that the apparent polar wander paths of the two continents have a similar history between ca. 1800 and 1730 Ma and between ca. 1650 and 1400 Ma. To achieve a fit within uncertainty limits, an additional relative latitudinal motion is necessary during the transition interval (1730–1650 Ma). We suggest that a two-stage evolution occurred between Laurentia and Australia during the assembly of the supercontinent Nuna. Australia and Laurentia traveled together between ca. 1800 Ma until ca. 1730 Ma in a semi-stable configuration. During 1730–1650 Ma, they underwent a net relative latitudinal motion, leading to the final assembly of Nuna after ca. 1650 Ma.

1. Introduction

Although some consensus has been reached concerning the configuration of the two youngest supercontinents Pangea (Torsvik et al., 2012) and Rodinia (Li et al., 2008; Evans, 2009), much more uncertainty pervades the internal organization of the older supercontinent Nuna (e.g., Evans and Mitchell, 2011; Zhang et al., 2012; Pisarevsky et al., 2014; Meert and Santosh, 2017). Recent work suggests that final Nuna assembly was not completed until ca. 1600 Ma (e.g., Pisarevsky et al., 2014; Nordsvan et al., 2018; Verbaas et al., 2018), and that the preceding ca. 2000–1700 Ma interval might represent an early assembly of the building blocks of Nuna including proto-Laurentia (e.g., Whitmeyer and Karlstrom, 2007; Hoffman, 2014), proto-Australia (e.g., Betts et al., 2016), and proto-Baltica (e.g., Bogdanova et al., 2013). Global paleogeography during this interval has been examined geologically (e.g., Betts et al., 2016; Zhao et al., 2002; Lubnina et al., 2017), but available paleomagnetic constraints are still not sufficient for establishing a robust paleogeography for the assembly of Nuna. Apart from a reasonably well-established core of Nuna containing Laurentia, Siberia, and Baltica (e.g., Wu et al., 2005; Wingate et al., 2009; Evans and Mitchell, 2011), another extensively studied connection is that between western Laurentia and eastern proto-Australia, where a complicated amalgamation history with assembling and rifting phases has been proposed between ca. 1880 and 1600 Ma (Betts et al., 2016). More recently, Meert and Santosh (2017) argued that the apparent polar wander paths of Laurentia and Australia between 1800 Ma and 1400 Ma are not in strong agreement after applying the reconstruction rotation parameters of Payne et al. (2009). Part of the reason for this discrepancy could be the presence of numerous large age gaps for which no high-quality paleomagnetic data are available for both Laurentia and Australia. For instance, between 1820 Ma and 1750 Ma, there is only one combined Elgee and Pentecost formations pole of ca. 1800 Ma(?)

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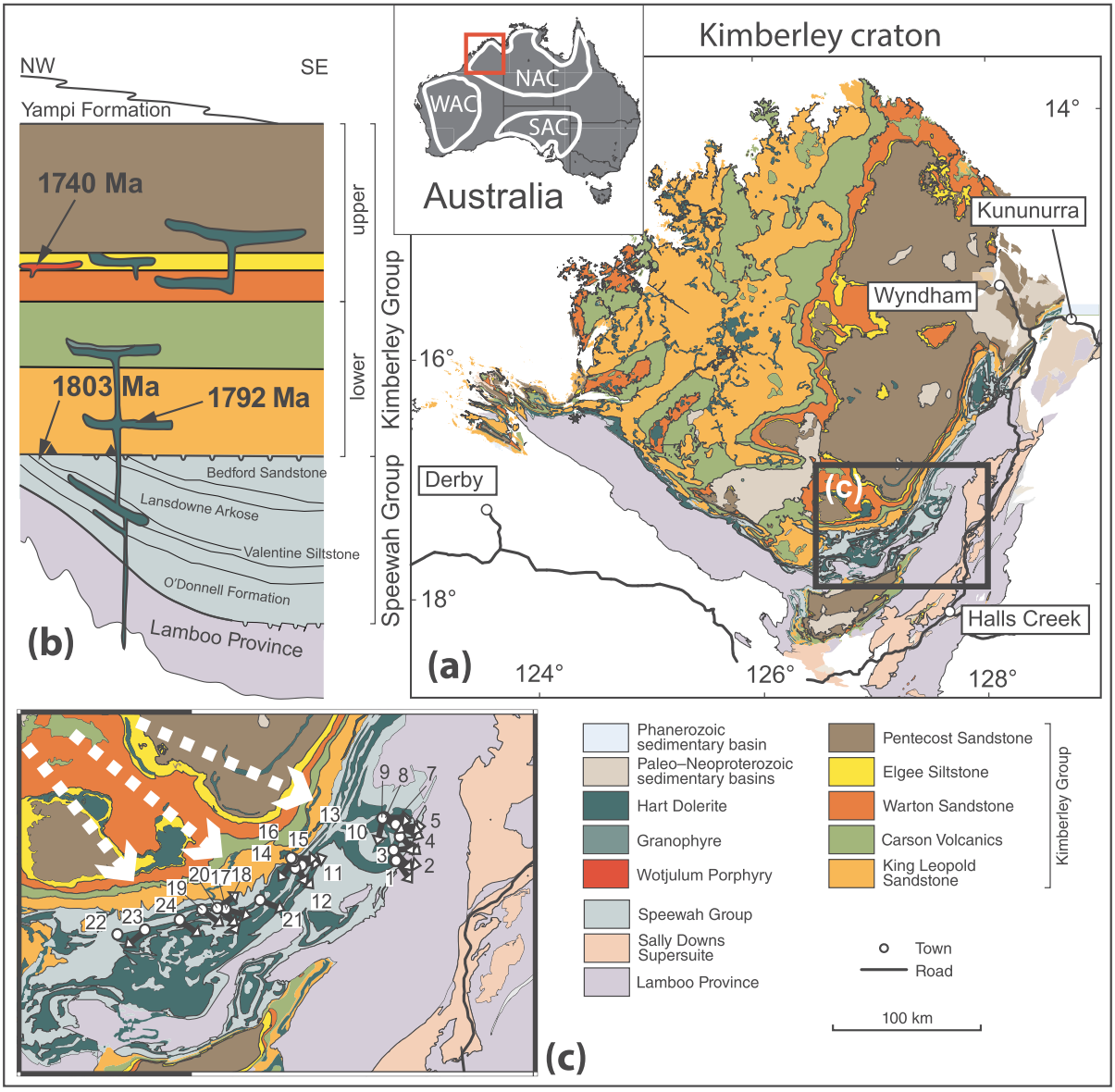


Fig. 1. (a) Geologic map of the study area, modified from Sheppard et al. (2012). (b) schematic stratigraphy of the Kimberley Group and intruding dolerite sills. Small inset at the top shows the position of the Kimberley craton relative to the Proterozoic cratons of Australia: the North (NAC), West (WAC) and South (SAC) Australian cratons. Main constraining ages in Ma are indicated (errors and explanations reported in the text). (c) Enlarged geologic map of the study area. Also shown are the paleomagnetic sampling sites from the Hart Dolerite and site mean direction of K1 axes of the AMS results (see text for further explanations). Large white dashed arrows are suggested directions of magma injection of the sills (see Sections 4.1 and 5.1).

for the Kimberley craton of northern Australia (Li, 2000; Schmidt et al., 2009), and the 1785 ± 35 Ma(?) Dubawnt Supergroup pole for Laurentia (Park et al., 1973). However, the Dubawnt Supergroup may in fact be considerably younger (~1750 Ma, Raub, 2008), and the age constraints for the sedimentary Elgee-Pentecost Formations are also poor. Additionally, there is no control on potential inclination shallowing for these sedimentary-derived poles (Kodama, 2012). A preliminary study of the Hart Dolerite from eastern Kimberley craton in northwestern Australia yielded very promising results (McElhinny and Evans, 1976), although a subsequent study by Schmidt and Williams (2008) was unsuccessful and these authors attributed their uninterpretable results to secondary alteration of the intrusive rocks. We present here a new study based on a more detailed sampling of the Hart Dolerite from the eastern Kimberley craton overlapping the field area of the original study of McElhinny and Evans (1976). We explore the paleogeographic implications of our resulting paleopole both in terms of the formation of the Australian continent and the assembly of the supercontinent Nuna.

2. Geologic setting

The Proterozoic strata of the Kimberley craton are dominated by two successive sedimentary basins, namely the Kimberley Basin and the underlying Speewah Basin. These basins are bound to the east and south by the ca. 1820 Ma Halls Creek Orogen (Sheppard et al., 2012); the orogen resulting from the collision of the Kimberley craton (KC) with the neighbouring terranes and the final assembly of the North Australian craton (NAC, Griffin et al., 2000; Cawood and Korsch, 2008). The Speewah Group is interpreted to have been deposited in a retro-arc foreland basin resulting from this orogen (Sheppard et al., 2012). Based on basin analysis, it has been suggested that the Kimberley Basin represents a semi-enclosed, shallow marine basin with sediment originating from the northwest (present-day coordinates, Gellatly et al., 1970; Sheppard et al., 2012).

The lower part of the Kimberly Group (King Leopold Sandstone and Carson Volcanics) is intruded by the voluminous Hart Dolerite sills and

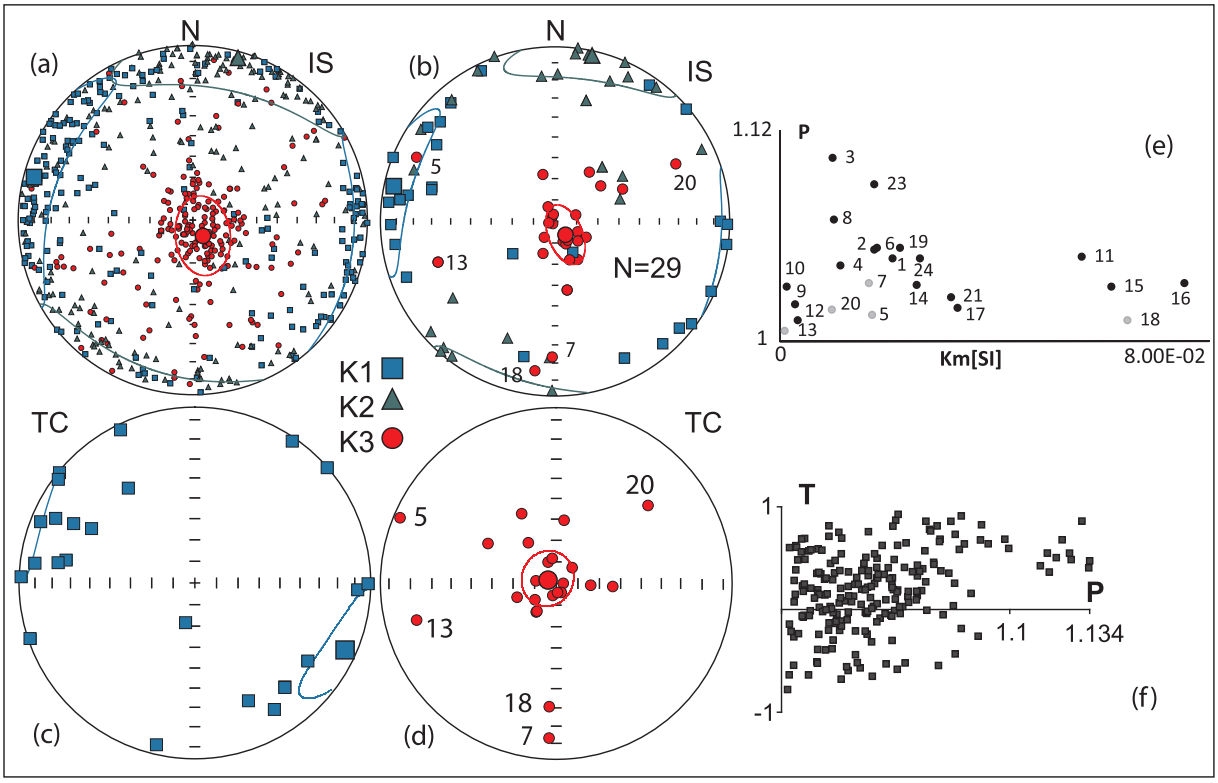


Fig. 2. Results of anisotropy of magnetic susceptibility (AMS) measurements. Stereonets show distributions of the maximum (K_1), intermediate (K_2) and minimum (K_3) axes of the anisotropy tensor for all individual measurements of 212 samples (a), site mean values (b) before tilt correction (In Situ) and site mean directions of only K_1 (c) and K_3 (d) after tilt correction (TC). Also shown are plots of degree of magnetic anisotropy (P) versus magnetic susceptibility (km) (e) and shape factor (T) versus degree of anisotropy (P) (f). Grey dots (e) indicate sites with an anomalous K_3 mean direction (d), due to their very low degrees of anisotropy. Number of these sites (5, 7, 13, 18, 20) are labeled in plots b and d. All other sites show a subhorizontal foliation (d).

associated dykes (Fig. 1). The Hart Dolerite has been interpreted as a post-collisional (post Halls Creek Orogeny) magmatic event caused by plate reorganization (Sheppard et al., 2012). Ion microprobe (SHRIMP) U-Pb analysis of the Hart Dolerite yields a mean age of 1797 ± 11 Ma (Sheppard et al., 2012), but more recently Ramsay et al. (2018) refined the age using the ID-TIMS U-Pb method on baddeleyite, with two upper intercept ages that overlap within uncertainty at 1792 Ma, though more precise $^{207}Pb/^{206}Pb$ ages and textural relationships within the gabbro intrusions indicate a slightly more prolonged magmatic history (~3 Myr) with multiple pulses of magma being emplaced in succession. The overlying upper Kimberley Group (Warton Sandstone, Pentecost Sandstone, and Elgee Siltstone) are intruded by unnamed dolerite sills that are undated, and their relationship with the Hart Dolerite is uncertain. The age ambiguity surrounding these dolerite sills introduces uncertainty regarding the ages of paleomagnetic poles obtained from the Elgee Siltstone and Pentecost Sandstone (Li, 2000; Schmidt and Williams, 2008). Based on geochemical similarities and the interpretation of recent geological reports for the area (Ramsay et al., 2017; Sheppard et al., 2012), we assume that those unnamed sills are coeval to the Hart Dolerite, which suggest an age of ca. 1803–1793 Ma for the Elgee-Pentecost pole. After the Halls Creek Orogeny, Mesoproterozoic, Neoproterozoic, and Paleozoic deformational episodes, namely the Yampi Orogeny, the King Leopold Orogeny and the Alice Springs Orogeny affected the Kimberley craton along its margins, particularly within the Paleozoic Canning and Ord basins (Tyler et al., 2012). In our study area, no significant post-Proterozoic folding has been reported (Derrick et al., 1968).

3. Paleomagnetic sampling and methods

A total of 232 oriented samples were collected from 31 sites of the Hart Dolerite (Fig. 1). Post-emplacement tilting of the sills has been obtained either through bedding measurements of adjacent sedimentary layers, or from the most recent geologic map (also based on adjacent sediments, Landsdowne and Dixon Range, Dow and Gemuts, 1967; Derrick et al., 1968). Paleomagnetic samples consist of oriented one-inch-diameter core samples drilled with a portable, water-cooled rockdrill with a diamond-mounted drill crown. Sample and strata orientations were measured using sun and magnetic compasses.

To obtain information about the magnetic minerals of the dolerite, a total of 12 representative samples were subjected to thermal demagnetization of a composite three-axis isothermal remanent magnetization (IRM) imparted with orthogonal fields of 2.50 T, 0.40 T, and 0.12 T (Lowrie, 1990). Additionally, the change of susceptibility as a function of temperature was obtained using an AGICO Kappabridge MFK-1 for representative samples of different lithology (fine to coarse grained). Using the same instrument, the anisotropy of magnetic susceptibility (AMS) was measured for one specimen from each sample prior to demagnetization. The results were analyzed using the Cureval8 and Anisoft 42 software provided by the same company. To retrieve the component structure of the Natural Remanent Magnetization (NRM), all 220 specimens were stepwise demagnetized using thermal (TH) and alternating field (AF) demagnetization techniques up to 600 °C or 110 mT, respectively. After each demagnetization step, the remanence was measured using an automated 2G SQUID cryogenic magnetometer (RAPID system Kirschvink et al., 2008). For comparison, a few samples were measured with an AGICO spinner magnetometer. AF demagnetization was performed using either the 2-G RAPID system or a tumbling AF demagnetizer. Thermal treatment was undertaken using an ASC thermal demagnetizer. To account for the presence of large amounts of multi-domain magnetite, which is common in coarse-grained mafic rocks, the samples were subjected to low-field AF demagnetization

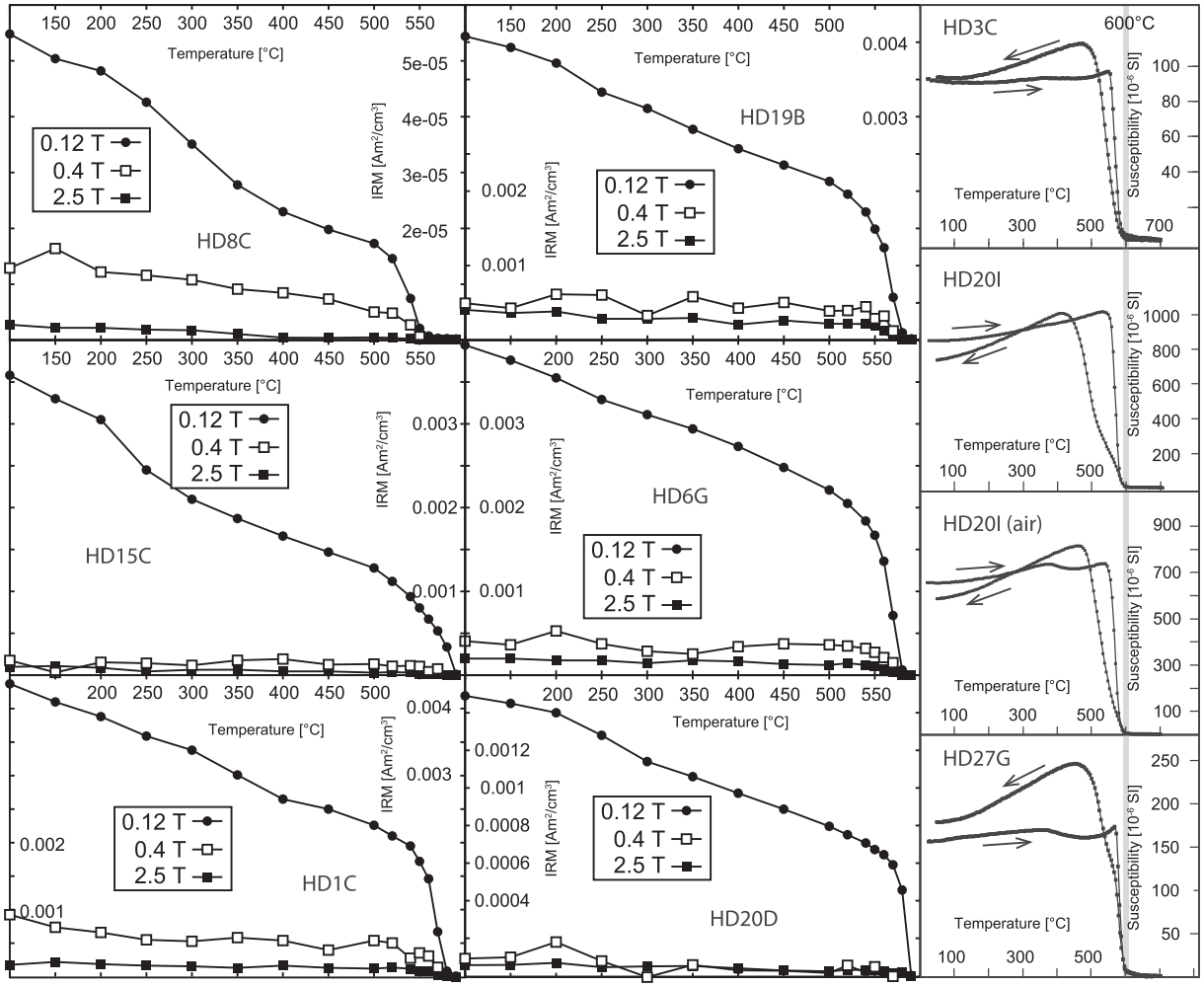


Fig. 3. Results of rock magnetic experiments on representative samples from the various sites of the Hart Dolerite. The six left panels show thermal demagnetization of orthogonal three axis IRMs (Lowrie, 1990) of 0.12, 0.40 and 2.50 T, respectively. The right small panels show results of susceptibility versus temperature experiments in argon atmosphere (HD20I was also performed in air).

(steps of 5, 10, 15, and 20 mT) prior to thermal treatment. For comparison, one set of 50 samples was treated with low temperature demagnetization (LTD) by bathing the samples in liquid nitrogen for approximately 10 min. All paleomagnetic measurements were carried out in a magnetically-shielded room at Curtin University in Perth. Analysis of the demagnetization results was performed on unidirectional segments of the demagnetization diagrams (Zijderveld et al., 1967) using the least-squares method of Kirschvink (1980) and combined with remagnetization great circles whenever necessary (McFadden and McElhinny, 1988). Mean directions were deduced from the individual directional data in combination with calculation of common statistical precision parameters (Fisher, 1953). Data analysis was conducted with software packages of 'PmagPy' version 3.4.0 (Tauxe et al., 2016) and 'PaleoMag' version 3.1b3 of Craig Jones. Paleopole analysis and paleogeographic reconstructions were conducted using Generic Mapping Tools Wessel and Smith (1998) and GPlates (Boyden et al., 2011).

4. Results

4.1. Anisotropy of magnetic susceptibility

The degree of AMS for the measured dolerite samples is generally moderate to low (P value of less than 1.10, Fig. 2), with only a few exceptions between 1.1 and 1.134, as typical of mafic intrusions (Ferré, 2002). Ferré (2002) obtained elevated P values from mafic intrusions and related them to the presence of pyrrhotite. Whereas the T value for the Hart Dolerite (HD) samples fit this scenario, the P values are too low to be indicative of pyrrhotite (Fig. 2, Ferré, 2002). A broad range of the magnetic fabric of oblate to prolate is present, with a slight majority of sites showing oblate fabric (Fig. 2).

Apart from few exceptions where the magnetic fabric data show very low degrees of anisotropy (Fig. 2), subhorizontal magnetic lineations (K_{MAX} or K_1 directions) are predominantly oriented in a NW-SE direction with a mean trend of 109° and not overlapping confidence ellipses of the K_1 and K_2 directions (using Jelinek statistics of mean K_1 directions, Jelinek, 1977, Fig. 2). Similar to previous studies on sills (Ferré, 2002; Lubnina et al., 2016), we interpret these results as being caused by alignment of magnetic particles parallel to the magmatic flow direction. This is in agreement with a general flow direction from the northwest in present day coordinates (Fig. 1c). The few exceptions to this general trend are all characterized by very low degrees of anisotropy (low P values, Fig. 2).

4.2. Rock magnetic results

Demagnetization of the composite three axis IRM (Lowrie, 1990) shows a general dominance of the low coercivity fraction in all samples (Fig. 3). Additionally, the magnetization is completely unblocked by heating to 600 °C. However, even while the 0.12 T fraction is the most dominant magnetization, the 0.40 T fraction is always present and

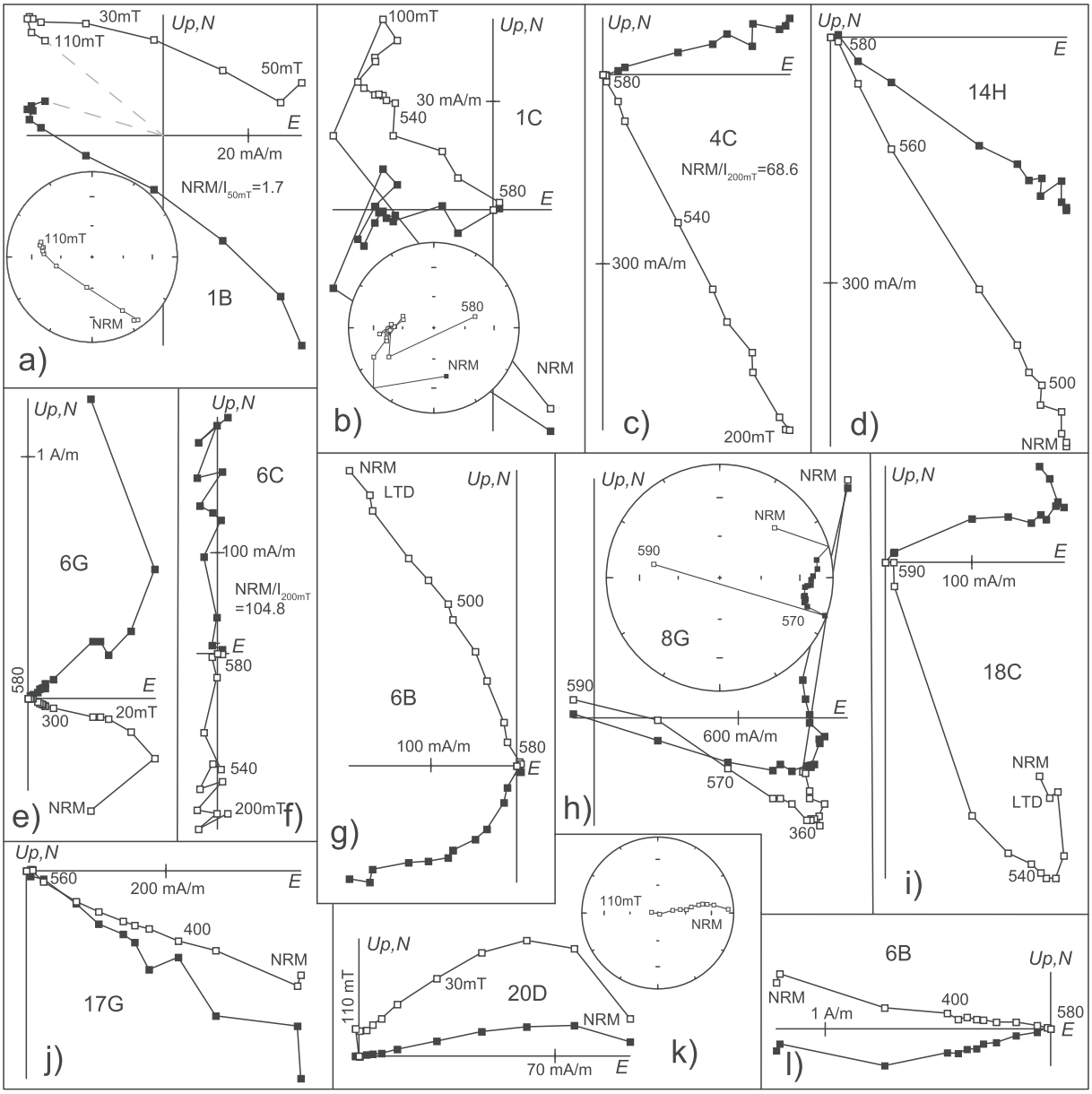


Fig. 4. Orthogonal vector end-point diagrams of thermal and alternating field demagnetization results of representative samples from all sites in tilt-corrected (stratigraphic) coordinates. Solid and open squares represent vector end points projected onto the horizontal and vertical planes, respectively. Demagnetization steps are given in °C, if not otherwise indicated.

shows similar behaviour during demagnetization (Fig. 3). This provides evidence for the existence of both single and multi-domain (SD and MD) magnetite in the mafic sills. There is a slight indication for another low temperature phase with an unblocking temperature of ~350 and 400°C (Fig. 3). This behavior may be related to the inversion of maghemite to hematite or the presence of pyrrhotite (Dunlop and Özdemir, 1997).

The susceptibility versus temperature curves are very consistent and show mainly a sharp drop towards a magnetite like Curie temperature and the presence of a slight Hopkinson peak (Fig. 3). The slight irreversible appearance of the curves might be related to a mixture of SD and MD magnetite as well as maghemite (Dunlop, 2014), and are therefore in agreement with our previous conclusions.

4.3. Paleomagnetic results

The NRM intensity of all measured samples is generally very strong (between hundreds and thousands of mA/m). However, after application of low field AF or LTD the intensity is significantly reduced, generally to more than 50%. The NRM intensity of each individual sample does not seems to be a factor in regards to the directional behaviour. Most samples show slight directional changes below 500 °C and an abrupt unidirectional decrease towards the origin between ~550 and 590°C (Fig. 4).

Whenever AF demagnetization was successful, the response agrees well with thermal demagnetization results (Fig. 4a, b). Also, initial LTD or low field AF demagnetization steps seem to result in similar behaviour (Fig. 4i). It is apparent, however, that demagnetization of some samples is not successful in separating a single characteristic remanent magnetization (ChRM) component (Fig. 4k). In such cases, remagnetization great circles were used in combination with fitted lines (McFadden and McElhinny, 1988) to obtain site mean directions (Fig. 4a, b, h, k). All but two sites yield either west-and-up or east-and-down directions (Fig. 5). Two sites (6, 14) contain samples that exhibit both upward and downward (dual-polarity) directions (Fig. 5). Apart

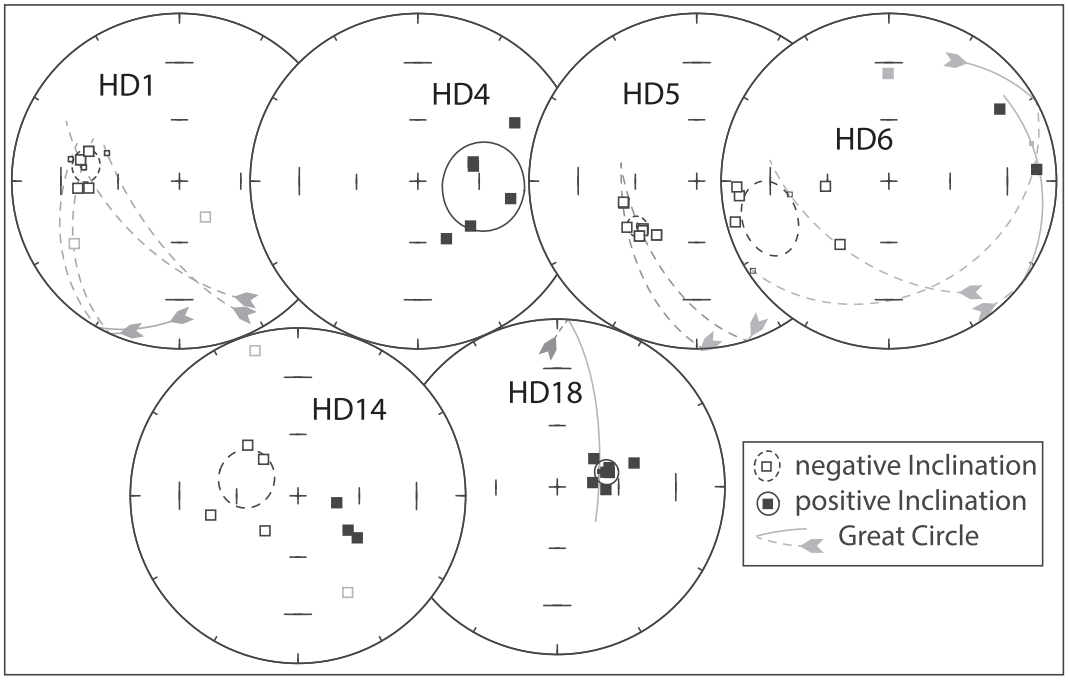


Fig. 5. Stereographic projections of site mean directions and confidence intervals (α_{95}) of six representative sites in stratigraphic (tilt-corrected, TC) coordinates. At sites HD1, HD5, HD6 and HD18, HTC component directions are combined with remagnetization great circles to obtain the site mean direction (McFadden and McElhinny, 1990).

Table 1
Site mean directions of sampled sites of the Hart Dolerite in geographic (in situ) and stratigraphic (tilt-corrected) coordinates.

Site	N/N★	#of	Bedding	In situ			Bedding-corrected		
		GC		D[°]	I[°]	α_{95} [°]	D[°]	I[°]	α_{95} [°]
HD1^{HQ}	8/14	4	323/51	280.4	−43.8	7.8	210.4	−58.0	7.8
HD2	7/10	0	323/51	88.5	16.3	24.3	62.6	37.6	24.3
HD3	5/13	2	284/29	252.6	−60.7	20.1	181.3	−74.8	20.1
HD4	6/8	0	260/20★	94.7	57.2	20.8	113.9	75.7	20.8
HD5^{HQ}	9/11	2	248/14	232.5	−53.2	5.0	224.4	−66.4	5.0
HD6^{HQ}	10/15	3	278/2	252.5	−25.9	15.7	252.1	−27.7	15.7
HD7	4/6	1	200/10★	107.6	73.8	18.6	139.1	71.4	18.6
HD8^{HQ}	17/20	6	198/18	72.3	23.9	14.9	80.8	33.4	14.9
HD9	2/7	1	195/18★	286.6	−47.0	17.8	305.1	−44.5	17.8
HD10^{EX}	9/9	5	195/18★	297.9	68.2	15.3	255.1	65.3	15.3
HD11	5/7	2	300/15★	63.5	27.8	21.1	55.4	35.2	21.1
HD12^{HQ}	5/6	2	300/15★	96.3	38.8	16.2	89.3	52.2	16.2
HD13^{HQ}	9/9	4	300/15★	76.6	37.1	14.2	66.4	47.1	14.2
HD14^{HQ}	7/9	0	300/15★	288.3	−63.4	13.9	274.7	−77.7	13.9
HD15^{HQ}	5/8	2	0/10★	150.8	26.5	14.8	147.7	35.1	14.8
HD16^{HQ}	9/9	6	0/10★	307.5	−75.8	10.0	263.8	−78.7	10.0
HD17	6/7	1	40/20★	312.2	−60.4	22.9	280.5	−55.4	22.9
HD18^{HQ}	7/7	0	40/20★	73.8	64.7	6.4	60.3	46.7	6.4
HD19	5/8	2	324/20	93.1	39.7	27.3	75.7	50.0	27.3
HD20	9/20	6	5/2★	297.5	−50.0	22.7	295.3	−50.7	22.7
HD21^{HQ}	6/8	3	334/9	108.9	6.1	11.3	107.9	12.4	11.3
HD22	5/5	1	330/12★	97.9	58.3	20.6	78.5	64.1	20.6
HD23	8/8	6	330/6★	280.5	−34.8	21.0	277.0	−38.6	21.0
HD24	5/9	2	330/8★	65.1	43.3	26.1	57.5	43.5	26.1
Mean									
Mean	23			92.6	47.4	10.0	83.8	55.3	10.0
Mean^{HQ}	11			92.4	45.1	19.4	82.4	52.9	17.2

Site: sampling site; N: number of samples (sites) used for calculating site (area) mean directions/number of measured samples; # of GC: number of great circles used for site mean directions (McFadden and McElhinny, 1990); Declination (D) and Inclination (I) in degrees; Fisher precision parameter k (Fisher, 1953); Fisher radius (α_{95}) of 95% confidence in geographic (*in situ*) and bedding corrected (TC) coordinates respectively; ★ bedding attitudes are based on nearest measurement of the GSWA; EX: site that is excluded from further analysis due to anomalous direction; HQ: sites with high quality results with α_{95} values below (or very close) to 16°.

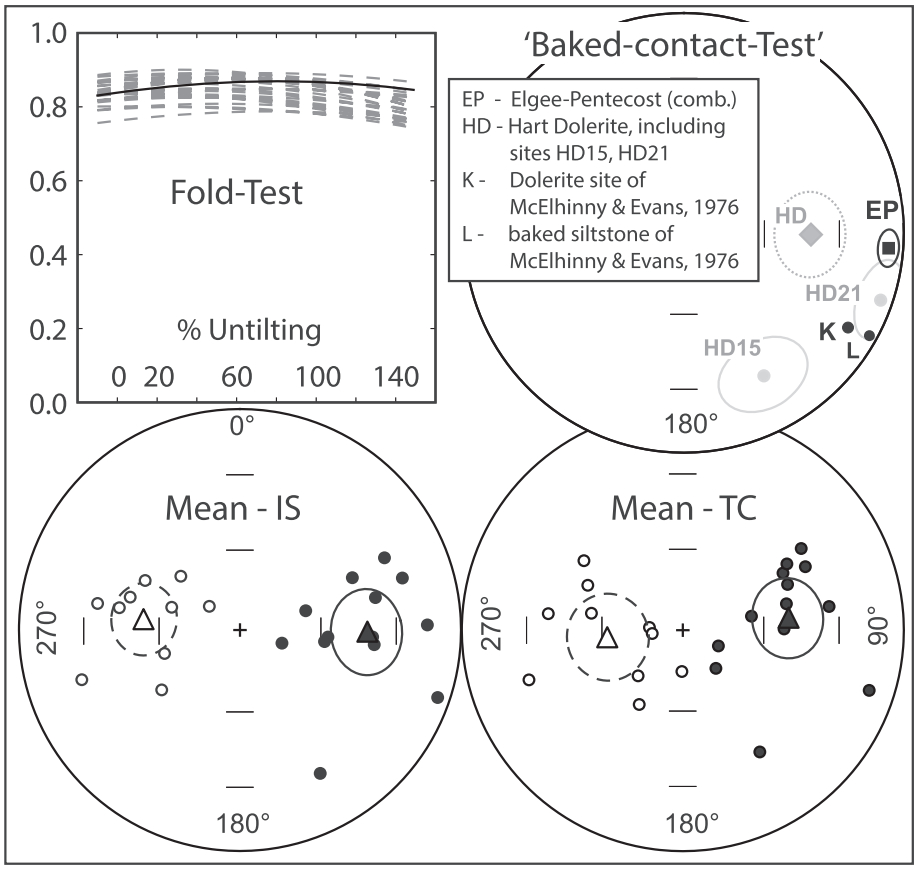


Fig. 6. Top panels show fold test and baked-contact test based on our results combined with the results of McElhinny and Evans (1976). Mean direction of the combined Elgee and Pentecost formations is based on Schmidt and Williams (2008). Fold test is shown using a non-parametric technique (Tauxe and Watson, 1994). Lower equal area projections show site and grand mean directions for positive and negative inclinations, respectively, in *in situ* (IS) and tilt-corrected (TC) coordinates.

from these exceptions, there are 10 sites with a mean negative inclination and 12 sites with a mean positive inclination. One site yields a mean direction after tilt correction that plots > 50° away from the overall mean and was excluded from further analysis ('EX' in Table 1). A total of 10 sites yield mean directions that have confidence intervals (α_{95}) that fulfill the common reliability criterion (#2) of Van der Voo (1990). The α_{95} of site 16 exceeds the limit of 16° by only 0.2° and was therefore included in the high-quality dataset. Overall mean directions were calculated for both the high-quality site mean directions (HQ, $\alpha_{95} < 16°$) and for the combination of all 23 site mean directions (after excluding only one outlier). The resulting mean directions before and after tilt correction for the whole and HQ site means cannot statistically be distinguished: an F test (Watson, 1956) yields values of 0.12 and 1.12 compared to the thresholds of 3.15 and 3.11 comparing the tilt-corrected distributions of all and the high-quality set, and for all the data comparing in situ and tilt-corrected distributions, respectively (Fig. 5).

We use the tilt-corrected mean directions of 23 sites to calculate an overall mean remanent direction of $D = 83.8°$, $I = 55.3°$ with $\alpha_{95} = 10.0°$, corresponding to a paleopole for the Hart Dolerite of *Plong* = 181.0°E and *Plat* = 5.3°S with a radius of 95% confidence $A_{95} = 12.0°$.

5. Discussion

5.1. Anisotropy of magnetic susceptibility

The K1 directions in geographic coordinates show a dominantly southeast pointing orientation (Fig. 1). Such a consistent trend implies that the potential magmatic source of the Hart Dolerite was likely located to the northwest in present-day coordinates, consistent with the observation that no similar sills are found farther east and south in the Kimberley craton. If the Hart Dolerite was indeed a product of a mantle plume (Rasmussen et al., 2012), the plume centre was likely located northwest of the study region, either on or off the Kimberley craton, or perhaps at its margin. We speculate that such a magmatic source location could be explained by a large igneous province being advected by edge convection (King and Anderson, 1998), as northwest of Kimberley approximates the margin of current reconstructions of supercontinent Nuna undergoing assembly at this age.

5.2. Reliability of paleomagnetic results

Our calculated overall mean direction for normal and reverse polarities are within 12° of being antipodal, constituting a positive reversal test (see above). We interpret this slight discrepancy as reflecting geomagnetic field reversal(s) that occurred during the timespan of magmatism. As pointed out by Ramsay et al. (2018), the Hart Dolerite represents a series of discrete pulses of igneous emplacement, which lasted over one million years, providing sufficient time for at least one field reversal to occur. We speculate that the anomalous HD10 site recorded a transitional field direction. The calculated confidence interval lies within predicted models of secular variation: the α_{95} of 10.0° compares favourably to the expected range of ~5° to 12° for such a paleolatitude (Deenen et al., 2011). Even if the scatter of the presented site mean directions is within expected values, the elevated error ellipses showing within site dispersion (α_{95} values in Table 1), might be related to the proposed geomagnetic field intensity low during this time, as speculated by McElhinny and Evans (1976). The time interval around 2 Ga has since been identified as the 'Proterozoic dipole low' (Shcherbakova et al., 2014; Valet et al., 2014), which is consistent with the assumptions made by McElhinny and Evans (1976).

Only two sites exhibit both polarities of remanence (Fig. 5), which might be related to the fact that sampled parts of certain outcrops without visible connection are actually parts of two separate intrusions. These observations suggest that distinct intrusion phases occurred,

Table 2
Selected paleopoles from Australia and Laurentia between 1440 and 1888 Ma.

Abbr.	Locality Name	Block	Age	±	λ	PLat	PLong	A_{95}	Q(6)	BD	REF
BP	Boot-Phantom Pluton	Lau.-T.-Huds.	1838	1	76.6	62.4	279.4	7.9	5	B	[1]
BP	Boot-Phantom Pluton	Lau.-T.-Huds.	1838	1	76.6	62.4	279.4	7.9	5	B	[1]
SD	Sparrow Dykes	Lau.-Rae	1827	4	31.7	12.0	291.0	7.9	5	B	[2]
Ma	Martin Formation	Lau.-Rae	1818	4	15.7	−9.0	288.0	8.5	5	A	[3]
Du	Dubawnt Group	Laurentia	1785	35	30.8	7.0	277.0	8.0	6	B	[4]
DP	Deschambault Pegmatites	Lau.-T.-Huds.	1766	5	74.5	67.5	276.0	7.7	4	B	[5]
JL	Jan Lake Granite	Lau.-T.-Huds.	1758	1	58.9	24.3	264.3	16.9	3	B	[6]
CD	Cleaver Dykes	Laurentia	1741	5	38.0	19.4	276.7	6.1	5	A	[7]
MB	Melville Bugt Dykes	Lau.-Green.	1633	5	16.8	5.0	273.8	8.7	5	B	[8]
WC	Western Channel Diabase	Laurentia	1590	3	30.8	9.0	245.0	6.6	4	A	[9]
StF	St.Francois Mountains	Laurentia	1476	16	20.6	−13.2	219.0	6.1	5	A	[10]
Mi	Michikamau Intrusion	Laurentia	1460	5	5.6	−1.5	217.5	4.7	5	A	[11]
Sp	Spokane Formation	Laurentia	1458	13	12.4	−24.8	215.5	4.7	5	A	[12]
Bo	Boonadgin Dykes (1)	WAC	1888	9	5.6	−47.0	235.0	6.0	7	A	[13]
Fr	Frere Formation (2)	WAC	1885		12.2	−45.2	220.0	2.0	6	B	[14]
PT	Plum Tree Volcanics (3)	NAC	1840	17		−29.0	195.0	14.0	2	D	[15]
HP2	Hamersley (4)	WAC	1795	25	10.0	−35.3	211.9	3.0	4	B-	[16]
HD	Hart Dolerite (5)	KC	1792	2	35.8	−5.3	181.0	12.0	7	A	this study
EP	Elgee-Pentecost combined (6)	KC	1769*	65	6.5	−5.4	211.8	3.2	4	A	[17]
PC	Peters Creek Volcanics (7)	NAC	1727	2	13.9	−26.0	221.0	4.8	6	A	[18]
Wo	Wollogorang Formation(8)	NAC	1727	4	11.9	−17.9	218.2	7.2	5	B	[19]
FC	Fiery Creek Formation (9)	NAC	1709	3	22.8	−23.9	211.8	10.4	4	B	[20]
WB	West Branch Volcanics (10)	NAC	1709	4	24.9	−15.9	200.5	11.3	5	A	[18]
Ma	Mallapunyah Formation (11)	NAC	1655	10	18.0	−35.0	214.3	3.1	6	A	[19]
To	Tooganinie Formation (12)	NAC	1648	3	30.9	−61.0	186.7	6.1	6	A	[19]
Em	Emmerugga Dolomite	NAC	1644	9	20.2	−79.1	202.6	6.1	5	B	[19]
Bl	Balbirini Dolomite, lower	NAC	1612	6	32.4	−66.1	177.5	5.7	5	B	[18]
Bu	Balbirini Dolomite, upper	NAC	1589	3	41.1	−52.0	176.1	7.5	5	B	[18]
GRV	Gawler Range Volcanics	SAC	1545	45	29.1	−60.4	50.0	6.2	3	B	[20]
MtI	Mt.Isa Metam.Dykes	NAC	1525	25	29.9	−79.0	110.6	8.4	4	B	[21]
BR	Blue Range, Pandurra formation	SAC	1440	140	29.3	−38.4	62.4	3.5	5	B	[22]

Abbr.: Abbreviation used in Fig. 8. Locality Name: For Balbirini Dolomite results from the lower and upper part are given. Numbers in brackets refer to pole numbers in Fig. 6. Block: Respective craton, Lau.-T.Huds.: Laurentia-Transhudson, WAC, NAC, SAC: West, North and South Australia Craton, KC: Kimberley Craton. Age: Proposed age with uncertainties. λ, PLat, PLong, A_{95}: Paleolatitude, Pole Latitude in °N, Pole Longitude in °E, and 95% confidence interval in °. Q(6): Paleomagnetic quality factor. BD: grade. REF: According reference: [1]: Symons et al. (1999); [2]: McGlynn et al. (1974); [3]: Evans and Bingham (1973); [4]: Park et al. (1973); [5]: Symons et al. (2000); [6]: Gala et al. (1995); [7]: Irving et al. (2004); [8]: Halls et al. (2011); [9]: Irving et al. (1972); [10]: Meert (2002); [11]: Emslie et al. (1976); [12]: Elston et al. (2002); [13]: Liu et al. (2018); [14]: Williams et al. (2004); [15]: Schmidt (2014); [16]: Li et al. (2000); [17]: Li (2000),Schmidt and Williams (2008); [18]: Idnurm (2000); [19]: Idnurm et al. (1995); [20]: Chamalaun and Dempsey (1978); [21]: Tanaka and Idnurm (1994); [22]: Schmidt and Williams (2011). *Previous age constraint, we follow: 1815 ± 20 Ma (see text for explanation).

which happened before and after a geomagnetic field reversal(s).

We observe similar confidence intervals (α_{95}) before and after tilt correction for the entire dataset, and a slight decrease in α_{95} for the HQ means (Table 1). The fold test on the entire data set is therefore indeterminate (Fig. 5, Tauxe and Watson, 1994). Accordingly, with the dataset available, it is not possible to determine whether the magnetization was acquired before or after folding. We note, nonetheless, that with the small amount of tilting, the eventual pre- and post-deformation directions are similar. The Hart Dolerite paleopole is distinct from the known Mesoproterozoic-Phanerozoic apparent polar wander path of Australia (Torsvik et al., 2012; Schmidt, 2014, Fig. 7), arguing against but not excluding subsequent remagnetization of the magnetic directions presented here (criterion #7 of Van der Voo (1990)). No baked contact test was conducted during this investigation. However, Fig. 3 from McElhinny and Evans (1976) shows the directions of a 'fresh' Hart Dolerite site and a 'baked' siltstone site, which was obtained from the Elgee Siltstone (Teronis member, Derrick et al., 1968). The directions of the fresh dolerite (site K) and the baked rocks (site L) are consistent and similar to some of our anomalous site mean directions (HD15 and HD21, Fig. 6). This result can be interpreted as a positive baked-contact test taking the results of Schmidt et al. (2009) into account (Fig. 6).

5.3. Paleoproterozoic paleomagnetism and the assembly of proto-Australia

To understand the assembly of proto-Australia during the Paleoproterozoic our results from the Hart Dolerite are compared with paleopoles from the NAC (including Kimberley), WAC and one overprint pole from the Hamersley province (Table 2). The other paleomagnetic result from the Kimberley craton is based on a combination of results from the Elgee Siltstone and Pentecost Sandstone (Fig. 1, Li, 2000; Schmidt and Williams, 2008). We used a correction of inclination shallowing on this combined pole assuming a flattening factor of $f = 0.6$ (e.g., Torsvik et al., 2012, 6^{SC} in Fig. 7). Inclination shallowing is still a rarely considered factor in Precambrian paleomagnetism of sedimentary rocks, but other studies suggest similar compaction-induced bias to the magnetic inclination as observed in the Phanerozoic (Kodama, 2012). More detailed studies of the Kimberley Group sedimentary rocks are needed to further verify the exact degree of inclination shallowing. There are a range of poles available from the McArthur Basin and the Lawn Hill Platform of the NAC, from 1727 to 1648 Ma (Idnurm et al., 1995; Idnurm, 2000; Schmidt, 2014). A ca. 1840 Ma pole from the McArthur Basin (the Plum Tree Volcanics pole, Idnurm et al., 1995; Schmidt, 2014) does not satisfy modern paleomagnetic reliability criteria and is excluded from further analyses.

The Elgee-Pentecost pole (Schmidt and Williams, 2008), corrected for inclination shallowing (flattening factor $f = 0.6$, e.g., Torsvik et al., 2012), is also used for our tectonic analysis. Most recent age constraints for the Capricorn Orogeny (peak deformation between 1820 and 1770 Ma, Johnson et al., 2013) and the Elgee-Pentecost strata (Sheppard et al., 2012) indicate a roughly coeval age for Elgee-Pentecost and Hamersley overprint (HP2) poles (Li, 2000, Table 2), whereas the Frere Formation pole is older (Rasmussen et al., 2012, Table 2). Comparison of the Hamersley overprint pole with the roughly coeval paleopoles of the Hart Dolerite and Pentecost-Elgee sedimentary rocks

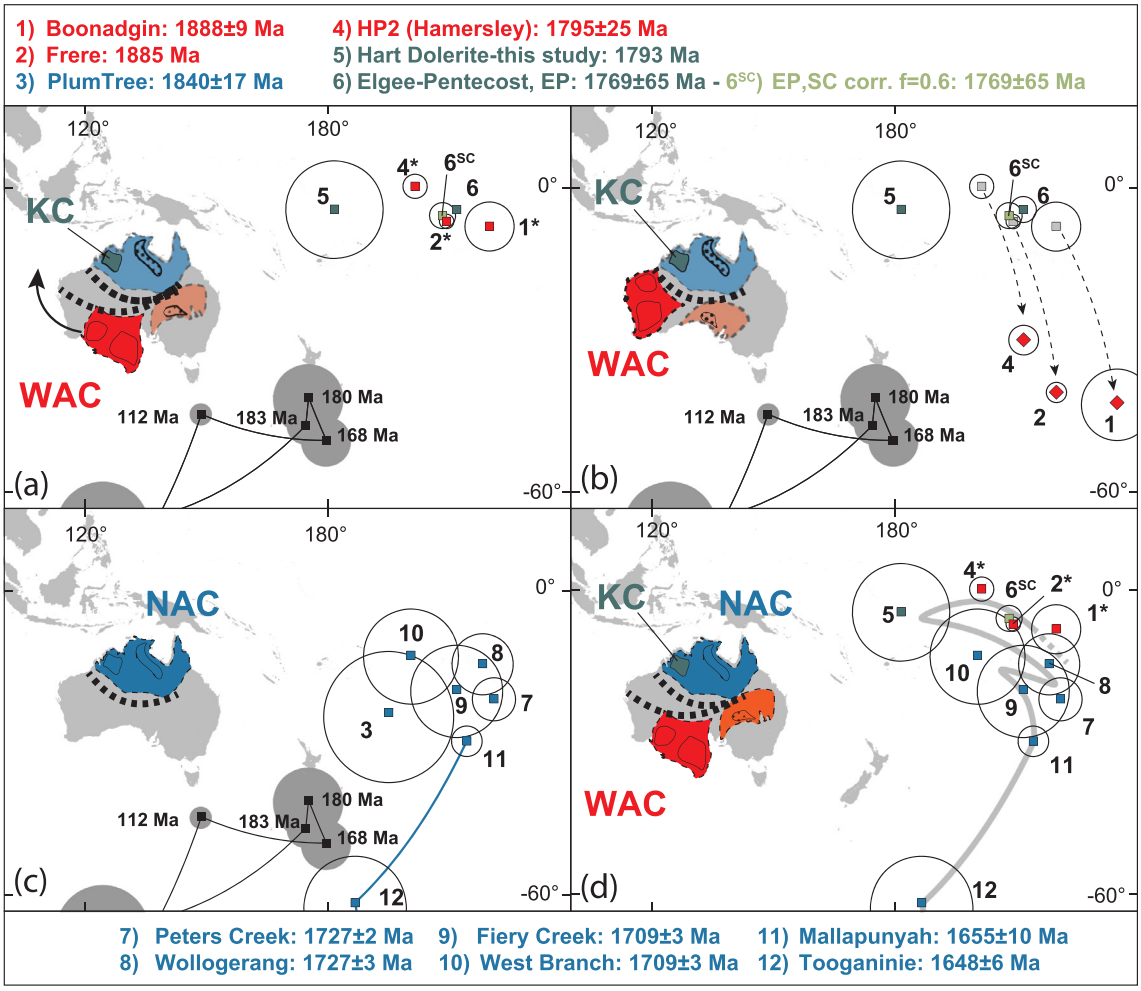


Fig. 7. Mercator projections of available high-quality paleopoles (Table 2) of the North (blue) and West (red) Australian cratons between 1640 and 1900 Ma including two results from the Kimberley craton (green). (a) Paleopoles of the Kimberley craton (KC) and West Australian craton (WAC) after correcting the intracontinental Neoproterozoic rotation (Li and Evans, 2011). (b) Same as (a) without intracontinental rotation. (c) Paleopoles of the North Australian craton (NAC) without KC poles. (d) Combination of all paleopoles yielding our proposed polar wander path for the combined proto-Australia (only WAC before 1800 Ma, poles 11–13). Pole-numbers are sorted according to age estimate. We include an inclination-corrected pole for Elgee-Pentecost (see text for explanation). We do not consider the paleopole from the Plum Tree Volcanics (#3, 1840 Ma, see text for explanation) as reliable. Only visible segment of the Phanerozoic APWP is shown within the projected area (Torsvik et al., 2012). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

shows a deviation of ~40° (Fig. 7). Applying an intracontinental relative clockwise rotation of 40° between the WAC and NAC (Li and Evans, 2011; Liu et al., 2018), a much better fit is achieved (Fig. 7a–b). This observation provides further support for the rotation model of Li and Evans (2011). Accordingly, all poles from the WAC were transferred into a NAC reference frame. Poles from the WAC are generally older in age and mostly predate the commonly assumed ~1.8 Ga assembly age for proto-Australia (Li, 2000; Williams et al., 2004; Pisarevsky et al., 2015; Liu et al., 2018).

The proximity of Australian paleopoles of ca. 1900–1700 Ma age range derived from different blocks (Fig. 7d) suggest that proto-Australia was assembled by that time, and that proto-Australia did not undergo large tectonic movements during this time interval. Furthermore, the previously suggested 'looping' apparent polar wander path (Idnurm et al., 1995; Li, 2000; Schmidt, 2014) is now relatively compressed and follows an oscillatory pattern (Fig. 7). Whether the observed APW oscillation is related to tectonic cycling (Betts et al., 2016) or true polar wander (Mitchell et al., 2010) requires further investigation.

5.4. Two-stage assembly of Australia and Laurentia in Nuna?

Using the rotation parameters from Payne et al. (2009) on NAC poles (and previously rotated WAC and SAC poles) transfers them into a Laurentian reference frame (31.5°N, 098°E, +102.5°; see also Evans and Mitchell, 2011), allowing for comparison of Australian and Laurentian APWPs. For NAC and WAC, the paleopoles discussed in the previous section were used, and compared with published high-quality poles for Laurentia (Table 2). Only a few pairs of coeval poles exist between Laurentia and proto-Australia for the age range between 1800 and 1300 Ma (Fig. 8). Following Payne et al. (2009), we match the Laurentian Cleaver Dyke pole with the Peters Creek and Wollogerang poles from Australia at ca. 1750 Ma (Fig. 8). From Laurentia, the pole based on the Dubawnt Supergroup could be coeval with the Hart Dolerite, however, it has a large age uncertainty and has previously been interpreted as a younger overprint (Raub, 2008). The 1770 Ma Deschambault Pegmatite pole (Symons et al., 2000) from Laurentia is about 20 Myr younger than the Hart Dolerite pole, but shows a similar excursion of APW as the Hart Dolerite pole (Table 2, Fig. 8). The quality of the Deschambault Pegmatite pole is not supported by field tests, and the pole plots very close to the present day spin axis. However, if the

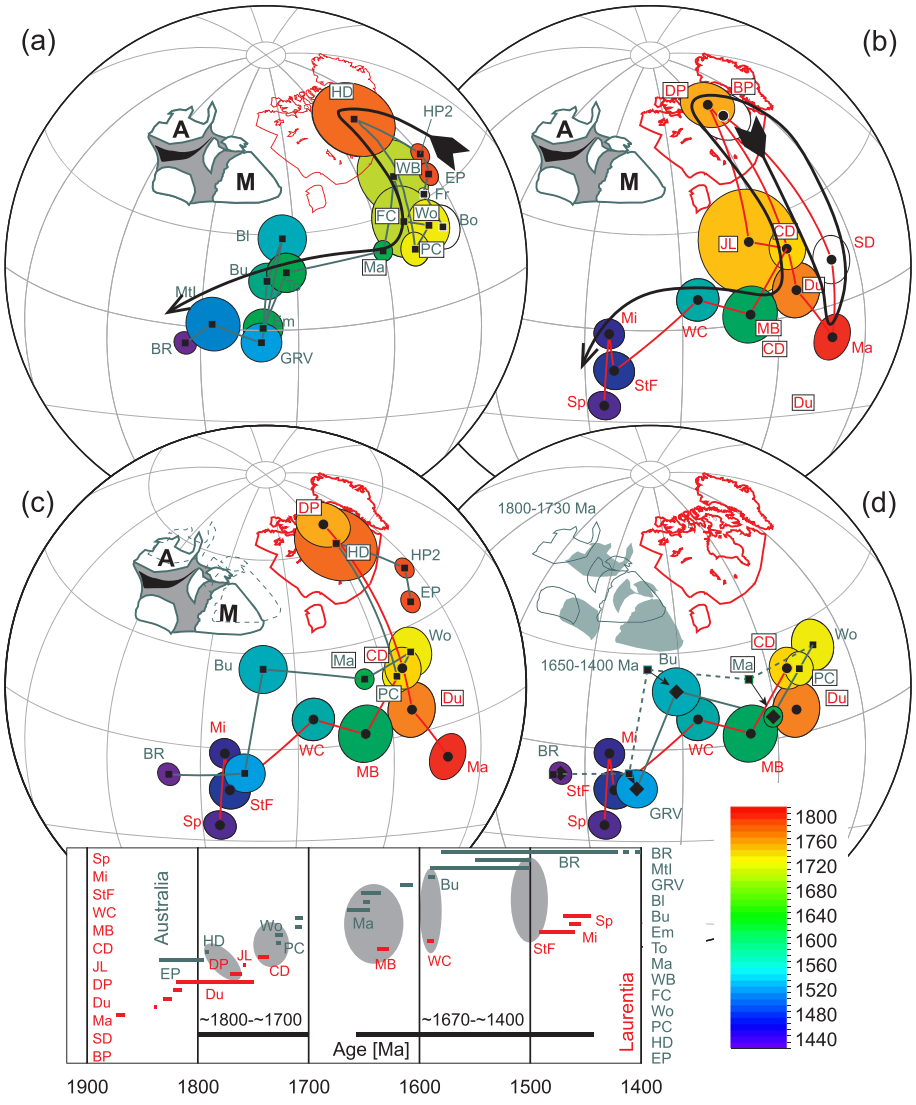


Fig. 8. Lambert equal-area projection of available high quality paleopoles for Australia and Laurentia between 1400 Ma and 1900 Ma (Table 2). (a) Paleopoles for Australia rotated to a common Laurentia reference frame using the Euler parameter (31.5°N, 098°E, +102.5°) of Payne et al. (2009). (b) Paleopoles for Laurentia. Black arrows in (a) and (b) indicate general trend. (c) Selected paleopoles of Laurentia (circles) and Australia (squares). (a-c) Outlines of NAC (A), WAC, SAC and Mawson (M) are also shown in the Laurentia fixed reference frame. (c) Outline of Australia and Mawson are shown according to Payne et al. (2009), dotted and using the best fit of our data. (d) Selected paleopoles of Australia with additional relative motion between 1750 and 1650 Ma according to a slightly modified Euler pole (37.8°N, 90.2°E, +102.7°) from panels a and c. Dotted line and squares without error intervals are before correction, diamonds and green line are after correction. Green shaded blocks represent NAC, WAC, SAC and Mawson after additional motion, green outlines before. (a-d) Error intervals of all paleopoles are color coded with age. Also a duration plot including error intervals of all shown paleopoles of Australia (green) and Laurentia (red) is shown. For abbreviations of paleopoles see Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

pole is of a primary origin, as the authors suggested (Symons et al., 2000), the paleomagnetic data would argue for a proximal position of proto-Australia to proto-Laurentia between ca. 1800 Ma and 1730 Ma (Fig. 8a–c). After ca. 1730 Ma, there is a consistent misfit apparent in at least two pairs of coeval poles from proto-Australia and Laurentia in that configuration (Fig. 8c). Rearranging proto-Australia and Laurentia with a different relative configuration between ca. 1650 and 1400 Ma (Fig. 8d) from before ca. 1730 Ma (Fig. 8c) accounts for this misfit. The paleomagnetic data suggest that proto-Australia and Laurentia moved coherently before ca. 1730 Ma and after ca. 1650 Ma (Fig. 8). Based on the data available, from ca. 1730 to ca. 1650 Ma, proto-Australia and Laurentia underwent a slight latitudinal shift in relative position.

Betts et al. (2016) proposed a two-stage amalgamation between the NAC and Laurentia for this period. Speculating that the two continents collided over the time interval between ca. 1900 and 1800 Ma during the Barramundi (NAC) and Wopmay (Laurentia) orogenies, the continents then drifted apart and re-collided at ca. 1600 Ma during the Isan (NAC) and Racklan (Laurentia) orogenies, thus, suggesting a full Wilson cycle occurred just prior to Nuna amalgamation. In this model, Australia and Laurentia are in a proto-SWEAT configuration (Payne et al., 2009). In the final collision, Laurentian terranes were transferred to the NAC, with subduction possibly occurring on both margins (Betts et al., 2016; Nordsvan et al., 2018).

Alternatively, paleogeographic reconstructions for this time period, primarily based on paleomagnetic data, argued that proto-Australia and Laurentia were not connected prior to ca. 1600 Ma using the existence of passive margin volcanism at 1685–1640 Ma and a mid-ocean ridge at 1650–1620 Ma between Laurentia and proto-Australia as evidence (Pisarevsky et al., 2014; Pehrsson et al., 2016). These models propose that there was an ocean between the continents until Nuna amalgamation at ca. 1600 Ma. We can account for this ocean and slightly modify the proposed reconstruction parameter of Payne et al. (2009) using an exact fit of our updated pole list (Hart Dolerite, Peters Creek, Wollogerang, Cleaver Dykes, Deschambault Pegmatite, Table 2, Fig. 8). This enlarges the available space between Laurentia and Australia for the 1800–1730 Ma interval (Fig. 8c). We note, however, that also the previous reconstruction of Payne et al. (2009) is permissible with the available data. Subduction off the margin of proto-Australia drove the closure of the ocean and culminated in the obduction of the Bonnetia Terrane during the Racklan Orogeny (Nielsen et al., 2013; Furlanetto et al., 2016; Thorkelsen and Laughton, 2016; Gibson and Holzschuh, 2017; Verbaas et al., 2018). In both the NAC and NW Laurentia, several scenarios have been proposed where dextral strike-slip motion occurred between the two continents (Southgate et al., 2000; Verbaas et al., 2018). Alternatively, other authors hypothesize that a Mediterranean type ocean existed between Australia and Laurentia between 1800 and ca. 1600 Ma (Pisarevsky et al., 2014, and references therein).

The two competing paleogeographic models during this period

diverge mainly regarding a rigid connection of Laurentia and proto-Australia between ca. 1800 and ca. 1730 Ma. The newly obtained Hart Dolerite pole is within uncertainty unable to differentiate between the different models. However, if the two continents were separated by an ocean basin between 1800 and 1730 Ma, they still show a similar sense of absolute motion, suggesting a pre-final Nuna amalgamation connection.

6. Conclusion

A new paleomagnetic study of the 1792 Ma Hart Dolerite in the Kimberley craton of northwest Australia reveals the presence of a well-grouped dual-polarity magnetization. The dissimilarity of the direction of the characteristic remanent magnetization from expected directions of younger ages for this region and a previously obtained baked contact test argue for a primary origin. The mean direction for 23 sites is $D = 83.8°$, $I = 55.3°$ with $\alpha_{95} = 10.0°$ after a minor tilt correction. This magnetic direction corresponds to a paleopole of $Plong = 181.0°$ and $Plat = -5.3°$ with an error ellipse of $A_{95} = 12.0°$ and a paleolatitude of ~36° (27.0–47.7°). A comparison of the obtained result with previous results from Australia and Laurentia yields the following implications: (1) the proto-Australian craton was assembled by around 1800 Ma; (2) the paleomagnetic results support the presence of a 40° intracontinental rotation within Australia after its Paleoproterozoic assembly; (3) there was a two-stage evolution between proto-Australia and Laurentia in the lead-up to the final assembly of the supercontinent Nuna: the first stage produced a semi-stable configuration between the continents for the 1800–1730 Ma interval while they moved coherently but were not arranged in the final position. Subsequently, a phase of oblique convergence (eventually following some minor rifting) occurred between the NAC and Laurentia during 1730–1650 Ma closing the small ocean basins between them. Such a two-stage assembly of Nuna is consistent with geologically-based models (e.g., Betts et al., 2016; Nordsvan et al., 2018).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, athttps://doi.org/10.1016/j.precamres.2018.12.026.

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